

PROBLEM SET 7: EXTRACTING THE BUSINESS CYCLE

I – CYCLES

Consider the two periodic macroeconomic variables:

$$\{X_t\}_{t=0}^{\infty} = \left\{ \sqrt{2} + a, -a, -\sqrt{2} + a, -a, \sqrt{2} + a, -a, -\sqrt{2} + a, -a, \dots \right\}, \quad (1)$$

$$\{X_t\}_{t=0}^{\infty} = \left\{ \sqrt{2} - a, a, -\sqrt{2} - a, a, \sqrt{2} - a, a, -\sqrt{2} - a, a, \dots \right\}. \quad (2)$$

- 1 – Draw a graph of the time evolution of X and Y .
- 2 – Compute the empirical mean, empirical variance and empirical covariance of those two processes.



Consider now the two following variables:

$$\{Z_t^1\}_{t=0}^{\infty} = \left\{ \sqrt{2}, 0, -\sqrt{2}, 0, \sqrt{2}, 0, -\sqrt{2}, 0, \dots \right\}, \quad (3)$$

$$\{Z_t^2\}_{t=0}^{\infty} = \left\{ a, -a, a, -a, a, -a, a, -a, a, -a, a, -a, \dots \right\}. \quad (4)$$

- 3 – Compute empirical means and variance of Z^1 and Z^2 as well as their empirical covariance.
- 4 – Show that for all i , Z^i is a periodic process. Compute period p^i and frequency f^i of each process.
- 5 – Compute the share of the empirical variance of X and Y that can be attributed to frequencies $1/2$ (high frequency) and $1/4$ (low frequency).
- 6 – Find meaningful ways of filtering X and Y in order to compute their empirical covariance at high and low frequencies.

II- EXTRACTING THE BUSINESS COMPONENT

Download quarterly data for GDP, Consumption, Investment and Hours worked for some country. I suggest you use the data web site of OECD (http://www.oecd-ilibrary.org/economics/data/main-economic-indicators_mei-data-en). Data are accessible if you are connecting from the university.

- 1 – Compute and plot business cycle components of those four variables using a band-pass filter (8,32) or the Hodrick-Prescott filter with smoothing parameter 1600. You might want to use or adapt the MATLAB programs `matlab_hpfilter.m` and `matlab_bpfilter.m` which are attached.
- 2 – Compute for each series standard deviation, standard deviation relatively to the s.d. of gdp, autocorrelation of order 1 and correlation with gdp.